

Kim Tuan Do

Department of Mathematics
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Academic Position

- 2025 **University of California, Santa Barbara**
Math Fellow
- 2022 – 2025 **University of California, Los Angeles**
Assistant Adjunct Professorship

Education

- 2016 – 2022 **Princeton University**
Ph.D. in Mathematics
Advisor: Christopher Skinner
Thesis: Construction of an anticyclotomic Euler system with applications
- 2015 – 2016 **University of Cambridge**
Master of Advanced Study in Mathematics
Graduated with Distinction
- 2012 – 2015 **Imperial College London**
BSc in Pure Mathematics
Graduated with 1st class honours

Research Interests

Arithmetic Geometry, Number Theory, Iwasawa Theory, Galois Representations, p -adic L -functions, Automorphic Forms, Arithmetic Langlands Program.

Awards and Achievements

- 2024 Liggett Instructor Distinguished Faculty Teaching Award in UCLA, awarded for valuable contributions to the teaching program.
- 2015 Governors' BSc Prize in Mathematics in Imperial College London, awarded to the best BSc in Mathematics at their final examination.
- 2014 Ken Allen Prize in Imperial College London, awarded for academic excellence.
- 2011 Bronze Medal in IMO 52nd, as member of the Vietnam team.

Pure Mathematics Publications & Preprints

- 2026 **K. T. Do**, *Euler system for self-dual Hecke character over biquadratic fields with applications*, preprint available at <https://kimtuando.github.io/website/BiES.pdf>
- 2026 **K. T. Do**, *Anticyclotomic Euler system over biquadratic fields*, **Algebra & Number Theory** 20 (2026), no. 8, 1615- 1647
- 2025 **K. T. Do** and F. Castella, *Diagonal cycles and anticyclotomic Iwasawa theory of modular forms*, **Journal of the European Mathematical Society (JEMS)**, online first available at <https://ems.press/journals/jems/articles/14299326>
- 2022 **K. T. Do**, *Construction of an anticyclotomic Euler system with applications*, **Ph.D. thesis, Princeton University**, available at <https://www.proquest.com/docview/2720964424>

Secondary Interdisciplinary Publications & Preprints

- 2026 N. A. Nghiem, **T. K. Do**, T.C. Wei, and T. V. Phan, *Quantum Algorithm for Estimating Ollivier-Ricci Curvature*, **Phys. Rev. Res.** 8, 023207, 2026, available at <https://journals.aps.org/prresearch/abstract/10.1103/hj52-lksk>
- 2025 N. A. Nghiem, **T. K. Do**, T. C. Wei, and T. V. Phan, *Quantum Algorithm for Estimating Intrinsic Geometry*, preprint available at <https://arxiv.org/abs/2508.06355>
- 2023 Ao, V.D., Tran, D.V., Pham, K.T., Nguyen, D.M., Tran, H.D., **Do, T.K.**, Do, V.H. and Phan, T.V. , *A Schrödinger Equation for Evolutionary Dynamics* Quantum Rep. 2023, 5, 659-682, available at <https://doi.org/10.3390/quantum5040042>
- 2022 L. T. Nguyen, **T. K. Do** and T. V. Phan, *On the Electrostatic Interaction Between Point Charges Due to Dielectrical Shielding*, Progress In Electromagnetics Research Letters, Vol. 107, 111-118, 2022, available at <https://doi:10.2528/PIERL22071401>
- 2022 **T. K. Do** and T. V. Phan, *Equal Radiation Frequencies from Different Transitions in the Non-Relativistic Quantum Mechanical Hydrogen Atom*, Quantum Rep. 2022, 4(3), 272-276, available at <https://doi.org/10.3390/quantum4030019>
- 2021 T. V. Phan, G. Wang, **T. K. Do**, I. G. Kevrekidis, S. Amend, E. Hammarlund, K. Pienta, J. Brown, L. Liu and R. H. Austin, *It doesn't always pay to be fit: success landscapes*, **J Biol Phys** (2021), available at <https://doi.org/10.1007/s10867-021-09589-2>
- 2019 T. V. Phan, R. Morris, M. E. Black, **T. K. Do**, K. Lin, K. Nagy, J. C. Sturm, J. Bos and R. H. Austin, *Bacterial route finding and collective escape in Mazes and Fractals*, **Phys. Rev. X** 10, 031017, available at <https://journals.aps.org/prx/abstract/10.1103/PhysRevX.10.031017>

Teaching experience

- 2025 – 2026 **Instructor** for Math 122A Complex Analysis
- 2024 – 2025 **Instructor** for Math 110B Group Theory (x2), Math 115A Linear Algebra (x2), Math 61 Discrete Maths (x1).
- 2023 – 2024 **Instructor** for Math 115A Linear Algebra (x3), Math 32B Multivariable Calculus (x2), Math 61 Discrete Maths. **Recipient of the Liggett Instructor Distinguished Faculty Teaching Award**
- 2022 – 2023 **Instructor** for Math 115A Linear Algebra (x2), Math 32B Multivariable Calculus (x2), Math 61 Discrete maths (x2)

2017 – 2022 **Teaching Assistance** at Princeton University: Foundations of Calculus; Multivariable Calculus (x2); Algebra I; Honors Linear Algebra; Advanced Linear Algebra with Applications; Numbers, Equations, and Proofs; Analytic Number Theory; Introduction to Riemann Surfaces; Complex Analysis with Applications

Invited Research Talks

- Dec 2024 'Euler systems', in the Institute of Mathematics of Vietnam Academy of Science and Technology, Algebra and Number Theory seminar
- Nov 2024 'Diagonal anticyclotomic Euler systems', in the 2024 Canadian Mathematical Society Winter Meeting
- Nov 2024 'Anticyclotomic Euler systems over imaginary quadratic and biquadratic fields', in the UCLA Number Theory seminar
- Sep 2023 'Construction of an anticyclotomic Euler system', in the UT Austin Number Theory seminar
- July 2023 'Construction of an anticyclotomic Euler system', in the Harish-Chandra Research Institute Arithmetic/Algebraic Geometry seminar
- Apr 2023 'Construction of an anticyclotomic Euler system', in the Caltech Number Theory seminar
- Nov 2022 'Construction of an anticyclotomic Euler system', in the UCSB Number Theory seminar
- Oct 2022 'Construction of an anticyclotomic Euler system', in the UCLA Number Theory seminar

Expository & Internal Seminars

- Oct 2024 'Gross Zagier formula', in the UCLA participating Number Theory seminar
- Dec 2023 'Cohomology of Shimura varieties and Matsushima's formula', in the UCLA participating Number Theory seminar
- Mar 2022 'Ihara lemma, anticyclotomic Euler systems for $V_f(\chi)$, and applications', in Automorphic forms and special values of L -functions class taught by Christopher Skinner
- Apr 2021 'Selberg Sieve - Λ^2 method', in Analytic Number Theory class taught by Peter Sarnak
- Oct 2020 'Classic arithmetic results about partition number $p(n)$ ', in Number Theory tea talk
- Dec 2019 'Overconvergent modular forms and Galois deformations', in IAS-Princeton University learning seminar about singular moduli for real quadratic fields led by Jan Vonk
- Oct 2019 'Introduction of theta correspondences', in a learning seminar on theta correspondences led by Gyujin Oh
- Apr 2019 'Mazur-Tate-Teitelbaum one-variable p -adic L -function and Fukaya-Kato two-variable p -adic L -function', in IAS-Princeton University learning seminar about Sharifi's conjecture led by Preston Wake
- Nov 2018 'Primitive roots and Erdos Moser Conjecture', in Numbers, Equations, and Proofs class taught by Christopher Skinner
- Mar 2015 'Erdős Moser Conjecture' in Undergraduate Colloquium of Imperial College London

Professional Service & Experience

- 2022 – 2025 **Organizer of the UCLA Number Theory Seminar**
- 2024 – 2025 Organizer of a working seminar about the Gross-Zagier formula